Direct two-stage routing

Choose the model before executing each stage.

RETRIEVAL ROUTER
Question +
candidate pool

→

RUN RETRIEVAL
Execute W or S only

→

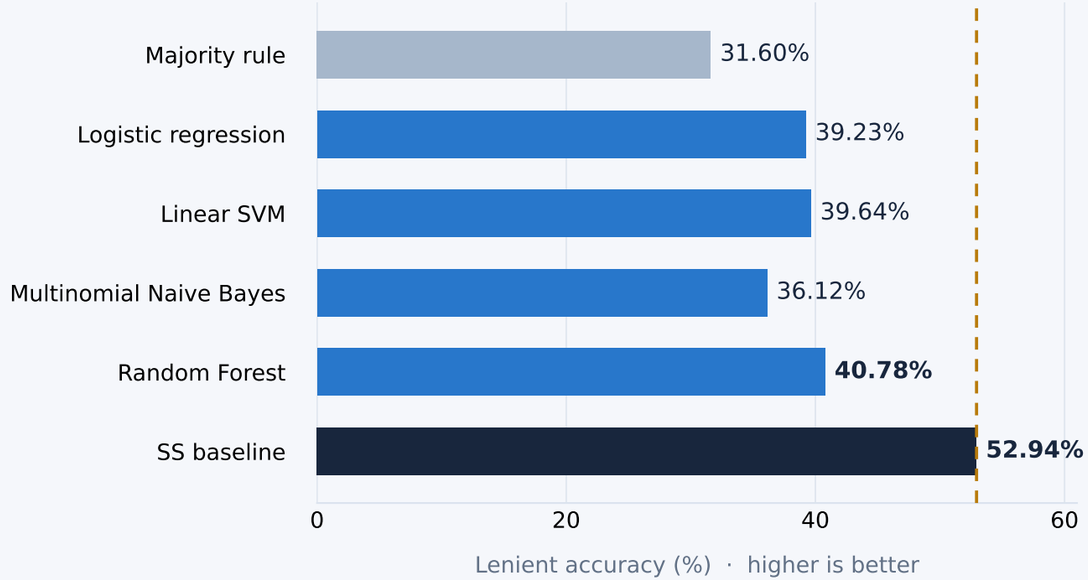
ANSWER ROUTER
Question +
chosen evidence

→

RUN ANSWERING
Execute W or S only

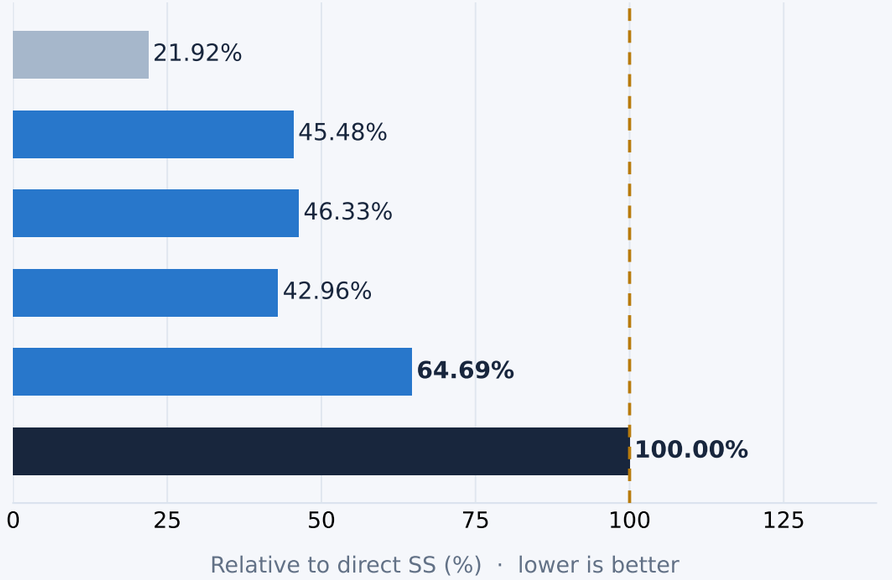
Each router chooses W/S before its stage executes. No weak answer is needed. SS executes exactly two model calls.

TEST ACCURACY



Lenient accuracy (%) · higher is better

PIPELINE COST



Relative to direct SS (%) · lower is better

Random Forest: 40.78% accuracy at 64.69% of SS cost.

GRPO has not been trained for this direct-routing method; cascade GRPO results are shown on page 2.

Same regenerated Instruct cohort | Test n = 1,481 | W = Qwen2.5-1.5B-Instruct | S = Qwen2.5-7B-Instruct

Selection: validation balanced accuracy. Costs include only the selected retrieval and answering calls.

Lenient scorer: review-validated reconstruction; full original scorer unavailable. Cost = model-size × recorded-token proxy; router overhead excluded.

SS is 52.94% on this test set; the original 52.79% belongs to a different validation cohort.

Weak-first cascade routing

Inspect the weak model’s output, then decide whether to escalate.

RUN WEAK RETRIEVAL
W selects references

→

RETRIEVAL REVIEW
Keep W or also run S

→

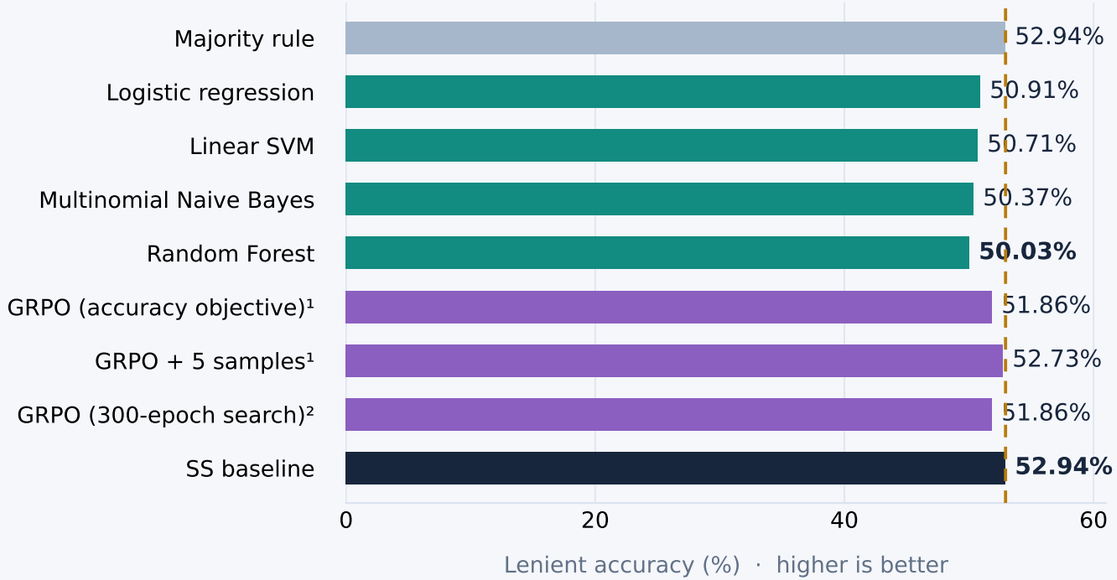
RUN WEAK ANSWERING
W generates an answer

→

ANSWER REVIEW
Keep W or also run S

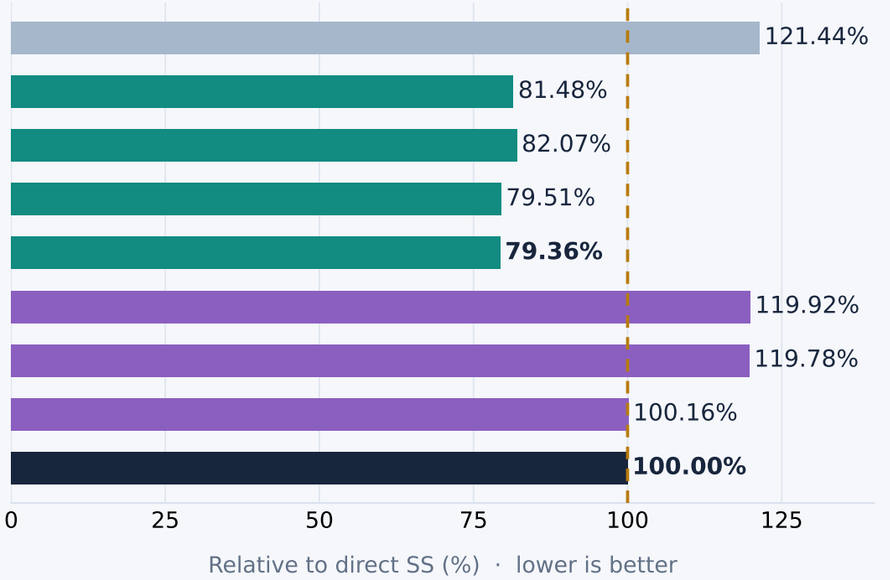
Retrieval review sees W-selected evidence; answer review sees evidence + the weak answer. Paid W calls are retained in cost.

TEST ACCURACY



Lenient accuracy (%) · higher is better

PIPELINE COST



Relative to direct SS (%) · lower is better

GRPO + 5 samples: 52.73% accuracy at 119.78% of SS cost; still below SS.

¹ Maximum-accuracy objective. ² Validation SS-budget objective; 300 epochs selected, test cost = 100.16%.

Same regenerated Instruct cohort | Test n = 1,481 | W = Qwen2.5-1.5B-Instruct | S = Qwen2.5-7B-Instruct

Classifiers: cheapest ≥95% of val SS. GRPO: three-seed linear heads; checkpoints/thresholds selected on validation. W calls + upgrades charged.

Lenient scorer: review-validated reconstruction; full original scorer unavailable. Cost = model-size × recorded-token proxy; router overhead excluded.

Purple = GRPO training extensions. Cascade SS can cost 121.44% of direct SS. Previously inspected test set; results are descriptive.